

Task 1 — Discover devices on your local network

Step 1: Check your Kali IP

- Run:
 - `ip addr`
- Look for something like:
 - `inet 192.168.29.167/24`
- If your network is:
 - `192.168.29.0/24`
- then your subnet is:
 - `192.168.29.0/24`

Step 2: Install Nmap

- On Kali:
 - `sudo apt update`
 - `sudo apt install nmap`
- Check the installation:
 - `nmap --version`

Step 3: Discover active devices

Use:

- `nmap -sn 192.168.29.0/24`

This creates:

- `scan_results.txt`

Step 4: View the file

Windows:

- type `scan_results.txt`

Kali/Linux:

- `cat scan_results.txt`

Task 2 — Reconnaissance Paragraph

Report paragraph

During the reconnaissance phase of a penetration test against an application such as Instagram, a tester would first collect information that is publicly available and permitted by the testing scope. This could include the application's official domains and subdomains, publicly exposed IP addresses, DNS records, technologies used by web services, publicly documented APIs, authentication mechanisms, and information about the application's externally visible attack surface. The tester would also identify the different services and entry points that are in scope for testing. Reconnaissance should be performed carefully and only against systems for which the tester has authorization.

Task 3 — Nmap scan of scanme.nmap.org

Run:

- `nmap -sS scanme.nmap.org`

If your Kali VM requires elevated privileges:

- `sudo nmap -sS scanme.nmap.org`

Save the output too

- `sudo nmap -sS scanme.nmap.org -oN scanme_results.txt`

Then:

- `cat scanme_results.txt`

Task 4 — Client-style penetration-testing report

****Basic Network Security Assessment Report****

A host-discovery scan was performed on the authorized local subnet to identify active devices.

The scan identified ****[X] active hosts**** out of the addresses examined.

A TCP SYN scan was also performed against the authorized Nmap test server, `scanme.nmap.org`.

The scan identified ****[X] open TCP ports****, including ****[list ports]****.

Open services represent potential attack surfaces and should be reviewed and securely configured.

No exploitation was performed during this assessment; the activity was limited to reconnaissance and scanning.

Further authorized testing would be required to determine whether any identified services contain exploitable vulnerabilities.